

An analytic counterexample to removing transversality from the local geometry of Schur idempotents

FA/Banach run `fa_banach_001`, lane 3

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Source metadata and status

The source is J. Parcet, M. de la Salle, and E. Tablate, *The local geometry of idempotent Schur multipliers*, arXiv:2312.02895v2, published in *Forum of Mathematics, Pi* 13 (2025), e14 [1]. Section 1.7 of the arXiv PDF (Section 2.7 in the published version) asks whether the equivalence in Theorem A survives without transversality when the second component of the boundary normal is nonzero. The authors expect counterexamples, but suggest that analytic boundaries probably do not admit them.

Claimed status: full counterexample, likely valid. Transversality is necessary even for a polynomial hypersurface in $\mathbb{R}^2 \times \mathbb{R}^2$.

1.7. On the transversality condition. The transversality assumption has been essential in our proofs of $\textcircled{1} \Rightarrow \textcircled{2} \Rightarrow \textcircled{3}$ in Theorem A, but it is not clear that it is really needed for the statement. Indeed, conditions $\textcircled{1}$ and $\textcircled{3}$ make sense without it, and $\textcircled{2}$ is already meaningful if one only assumes that $\mathbf{n}_2(x_0, y_0) \neq 0$, and we do not have an example where the equivalence fails. It is likely that such examples can be found, but probably not for domains with analytic boundary. We leave these questions as open problems. In the degenerate case where $\mathbf{n}_1$ is identically 0, or equivalently when Σ is locally of the form $\Sigma = \{(x, y) : y \in \Omega\}$, all conditions in Theorem A hold. The S_p -boundedness is in that case even true for $1 \leq p \leq \infty$ because the Schur multiplier whose symbol is the indicator function of Σ is just the right-multiplication by the orthogonal projection on $L_2(\Omega)$.

2. Idempotent Fourier multipliers on Lie groups

Proof intuition

Make many boundary sections pass through one degenerate fiber while their tangent directions vary. Ordinarily such a continuum of directions would trigger the paper’s Meyer–Fefferman obstruction. Here every meeting point has vanishing first normal component, so that blow-up cannot see it.

Boundedness comes from a different feature. The sign of one y -coordinate splits the symbol into just two pieces. On one piece it is the triangular projection, and on the other it is the same projection with the order reversed. Thus a changing analytic family of boundary lines is hidden inside a finite measurable pasting of familiar bounded multipliers.

Counterexample theorem

Theorem 1. *Let $U = V = (-1, 1)^2$, write $x = (r, q) \in U$ and $y = (s, t) \in V$, and put*

$$\Sigma = \{(x, y) \in U \times V : s > r^2 t\}.$$

For every $1 < p < \infty$, the indicator $\mathbf{1}_\Sigma$ is a bounded Schur multiplier on

$$S_p(L_2(U), L_2(V)).$$

At $z_0 = ((0, 0), (0, 0)) \in \partial\Sigma$, the V -component of the boundary normal is nonzero, but the zero-curvature condition (2) and the triangular representation (3) of Theorem A both fail on every neighborhood of z_0 . The boundary is real analytic.

Proof

Define

$$F(r, q, s, t) = s - r^2 t.$$

Then $\Sigma = \{F > 0\}$ and

$$\nabla F = (-2rt, 0, 1, -r^2).$$

In particular, ∇F never vanishes, so $\partial\Sigma$ is a real-analytic hypersurface. At z_0 its two normal components are

$$\mathbf{n}_1(z_0) = (0, 0), \quad \mathbf{n}_2(z_0) = (1, 0).$$

Thus z_0 is nontransverse but satisfies precisely the weaker normal hypothesis proposed in the source question.

We next prove boundedness. Let

$$V_+ = \{(s, t) \in V : t > 0\}, \quad V_- = \{(s, t) \in V : t < 0\},$$

and define measurable scalar maps

$$a(r, q) = r^2, \quad b(s, t) = \frac{s}{t} \quad (t \neq 0).$$

If $\tau_<(u, v) = \mathbf{1}_{\{u < v\}}$ and $\tau_>(u, v) = \mathbf{1}_{\{u > v\}}$, then, outside the null set $t = 0$,

$$\mathbf{1}_\Sigma(x, y) = \mathbf{1}_{V_+}(y) \tau_<(a(x), b(y)) + \mathbf{1}_{V_-}(y) \tau_>(a(x), b(y)). \quad (1)$$

The symbols $\tau_<$ and $\tau_>$ are the continuous triangular and reverse-triangular Schur projections. They are bounded on S_p for every $1 < p < \infty$; denote a common bound by C_p .

The pushforward of Lebesgue measure on U under a is absolutely continuous with respect to Lebesgue measure on $\mathbb{R}$ (on $(0, 1)$ its density is a constant multiple of $u^{-1/2}$). The pushforwards of Lebesgue measure on V_+ and V_- under b are also absolutely continuous: for a null set $A \subset \mathbb{R}$, Fubini's theorem gives

$$|\{(s, t) \in V_\pm : s/t \in A\}| = 0.$$

The source's change-of-variables lemma [1, Lemma 1.1] therefore gives

$$\|\tau_< \circ (a \times b)\|_{MS_p(U, V_+)} \leq C_p, \quad \|\tau_> \circ (a \times b)\|_{MS_p(U, V_-)} \leq C_p.$$

Restriction to V_+ or V_- is left multiplication by an orthogonal projection on $L_2(V)$ and hence contractive on every Schatten class. Equation (1) now yields

$$\|\mathbf{1}_\Sigma\|_{MS_p(U, V)} \leq 2C_p < \infty.$$

It remains to check that the other two conditions fail. For fixed $x = (r, q)$,

$$\partial\Sigma_x = \{(s, t) : s = r^2t\}, \quad T_{(0,0)}\partial\Sigma_x = \text{span}\{(r^2, 1)\}.$$

Every section passes through $y_0 = (0, 0)$, but these tangent lines vary with r . Choosing $r = 0$ and any sufficiently small $r \neq 0$ disproves the zero-curvature condition on every product neighborhood of z_0 .

For completeness, suppose there were neighborhoods U_0, V_0 and C^1 functions $f_1 : U_0 \rightarrow \mathbb{R}, f_2 : V_0 \rightarrow \mathbb{R}$ such that

$$\Sigma \cap (U_0 \times V_0) = \{(x, y) : f_1(x) > f_2(y)\}.$$

Since (x, y_0) is a boundary point for every $x \in U_0$, continuity forces $f_1(x) = f_2(y_0)$ for every such x . Hence the set on the right is independent of x . This is impossible: choose small $\rho, t > 0$ with $x^0 = (0, 0), x^\rho = (\rho, 0) \in U_0$ and $y = (\rho^2t/2, t) \in V_0$. Then $(x^0, y) \in \Sigma$ while $(x^\rho, y) \notin \Sigma$. Thus condition (3) also fails. $\square$

Scope, novelty check, and review recommendation

The example gives a negative answer to both parts of the source question: the equivalence fails without transversality, and it can fail for an analytic boundary. Both factor manifolds have dimension two. The result makes no claim about the paper's separate global questions for Fourier multipliers on $\text{SL}_2(\mathbb{R})$.

Run-index and bounded web searches through August 13, 2026 used the arXiv id, the exact transversality paragraph, analytic-boundary variants, and the formula $s > x^2t$. They found the 2025 publication and a 2025 thesis still presenting the transverse theorem, but no later solution or matching counterexample. This is a bounded novelty audit, not a priority claim.

Human review recommendation. Review as a likely valid full counterexample. The key audit points are the two-piece identity (1), applicability of the source's pullback lemma to a and b , and the boundary-point argument excluding a single triangular model.

References

- [1] J. Parcet, M. de la Salle, and E. Tablate, *The local geometry of idempotent Schur multipliers*, arXiv:2312.02895v2; Forum Math. Pi **13** (2025), e14, doi:10.1017/fmp.2025.6.